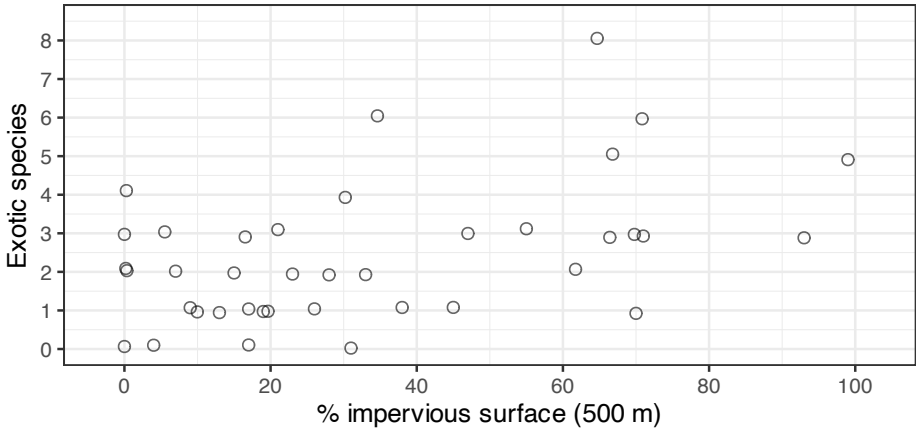
Exotic bees and Poisson regression (answer key)

Does urbanization favor exotic bees? Fitch et al. (2019) surveyed bees at 41 gardens and natural areas around Ann Arbor, Michigan. Each point is one site: the percent of the land within 500 m covered by impervious surfaces (roads, roofs, parking lots), and the number of exotic bee species found there (exotic species richness).



$$\log(\lambda) = \text{the natural log of } \lambda \qquad \lambda = e^{\log(\lambda)}$$

- λ: the expected (mean) count, greater than 0
- $\log(\lambda)$: any value from $-\infty$ to ∞

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1. Write the model

Write the model in statistical notation. What's the distribution of the response? What links λ to the linear equation?

$$\begin{aligned} \text{ExoticRichness} &\sim \text{Poisson}(\lambda) \\ \log(\lambda) &= \beta_0 + \beta_1 \text{PctImpervious} \end{aligned}$$

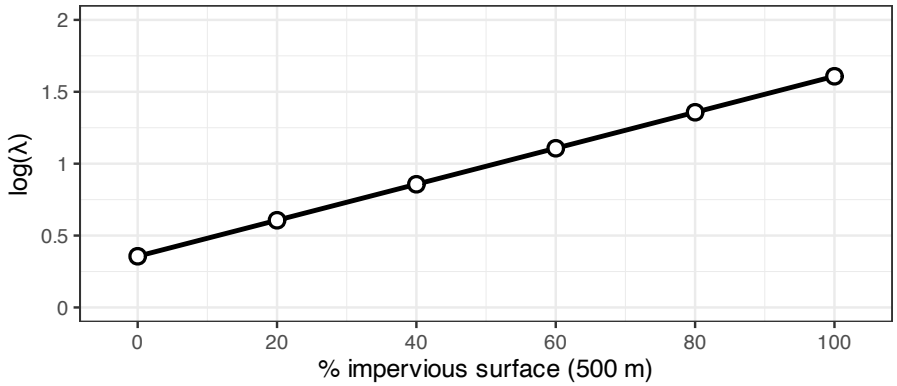
2. Draw the linear part

We fit the model in R:

```
bee_glm <- glm(exotic.richness ~ impervious, family = poisson(link = "log"), data = bees)
```

Term	Coefficient	Estimate
`(Intercept)`	β_0	0.356
`impervious`	β_1	0.0125

Draw $\log(\lambda) = \beta_0 + \beta_1 \text{PctImpervious}$. It's a straight line, so you only need two points.



3. Calculate λ

For each value of impervious surface, calculate $\log(\lambda)$ from the coefficients. Then convert it to the expected number of exotic species, λ.

% impervious	$\log(\lambda) = \beta_0 + \beta_1 \times$ impervious	$\lambda = e^{\log(\lambda)}$
0	0.36	1.43
20	0.61	1.83
40	0.86	2.36
60	1.11	3.03
80	1.36	3.89
100	1.61	4.99

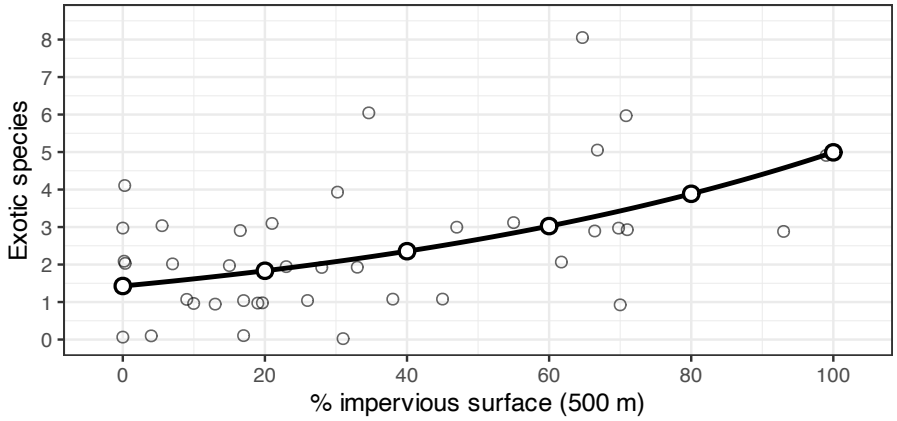
How much does $\log(\lambda)$ change between each row? 0.25

Divide each λ by the one above it. What do you get? 1.28 every time ($e^{20\beta_1}$)

In words, what happens to the expected number of exotic species for every 20 points of impervious surface? It's multiplied by 1.28, a 28% increase. Adding a constant on the log scale is multiplying by a constant on the count scale.

4. Draw λ

Plot your six values of λ on top of the data, then connect them with a smooth curve.



5. Is a Poisson a good fit?

A Poisson distribution's variance equals its mean. That's about a *given* value of the predictor, so we compare sites with similar impervious surface.

% impervious	Sites	Mean exotic species	Variance
0–25%	20	1.60	1.41
25–50%	10	2.00	3.56
50–100%	11	3.82	3.96

Is the Poisson a reasonable model for these data? Explain using the table.

Yes, reasonably. Within bins of impervious surface, the variance is close to the mean in the low and high bins. The middle bin's variance is higher, but it has only 10 sites, so its variance is noisy. Keep this check in mind for the drinking water violations.

Data: Fitch, G., et al. (2019). Does urbanization favour exotic bee species? Implications for the conservation of native bees in cities. *Biology Letters*, 15, 20190574. Adapted from Giacomangeli, G. (2023). Does Urbanization Favor Exotic Species? *QUBES Educational Resources* (CC BY-SA 4.0).